KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY
COLLEGE OF ENGINEERING
DEPARTMENT OF AGRICULTURAL ENGINEERING

AE 451 IRRIGATION AND DRAINAGE

CLASS ASSIGNMENT

Problem: Infiltration data obtained from a field measurement using the double ring infiltrometer is provided in the table below.

SN	Time (*t*, min)	Infiltration rate (*i*, mm/min)
1	1	3.80
2	2	2.39
3	3	1.68
4	4	1.63
5	9	1.38
6	14	1.18
7	19	1.10
8	29	0.98
9	39	0.88
10	49	0.84
11	60	0.80
12	80	0.68
13	100	0.67
14	120	0.63

1. Using the Microsoft Excel Software, plot the following curves:
    - infiltration rate (*i*, mm/min) against time (*t*, min)
    - *i* (mm/min) against $t^{-0.5}$ ($min^{-0.5}$)
2. Fit the curves with an appropriate trend lines, generating the *Kostiakov and Philips* equations and their r^2 values.
3. Determine the infiltration rate after an elapsed time of 3 hours from the *Kostiakov* equation.
4. Derive the cumulative infiltration (*d*, mm) versus time (*t*, min) equation from the *Kostiakov* equation obtained in step 2 above.
5. Determine the sorptivity (m $d^{-1/2}$) and the saturated hydraulic conductivity (m d^{-1}) of the soil from the *Philips* equation.
6. How long will it take for 75 mm of irrigation water to be infiltrated into this soil (h)?